2026-10-16 02:00Z

SPINNER VS SPINNER DOUBLE HIT

PASS VERIFIED

Does blade bending make a vertical spinner and a horizontal spinner hit twice in one engagement?

DOUBLE HIT REPRODUCED? PASS

YES

fast double hits: H bending 19%, both bending 5%, rigid 1%

No failure modes defined: ask the user what failure means for this part.

SPINNER VS SPINNER DOUBLE HIT

Time histories of the governing states (nominal case)

Blade-to-blade contact force



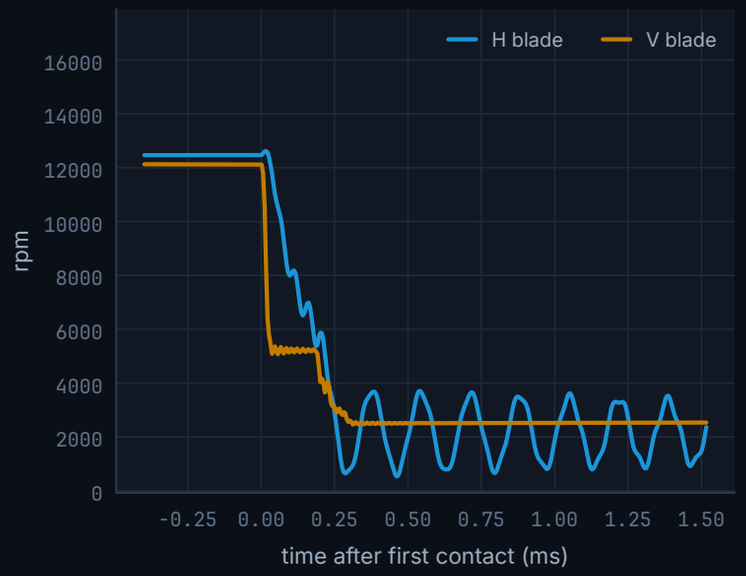
H tooth: in-plane (lag) bend



Out-of-plane (flap) tip deflection



Blade speed



2026-10-16 02:00Z

SPINNER VS SPINNER DOUBLE HIT

What to design around: drivers of the governing margin and the design-variable sweep

SIMBLOCKS // VERIFICATION

2026-10-16 02:00Z

SPINNER VS SPINNER DOUBLE HIT

PASS VERIFIED

Automated verification battery

CHECK	STATUS	DETAIL
determinism	PASS	bit-identical repeat
finite_outputs	PASS	all outputs finite
physical_bounds	PASS	5 bounds respected
stability_margin	PASS	critical dt 4.2e-06 s / dt 6.25e-08 s = 67.2x
dt_convergence	WARN	fast_double: err 0% (order inf); J1_Ns: err 0.017% (order 2.2); J2_Ns: err 0.23% (order 2); gap12_ms: err 0.11% (order 3.6); n_hits: err 0% (order ...
energy_ledger	PASS	worst closure residual 0.086% across 3 dt levels (tol 5%)
limit:robots never meet (1 ms run) → no hits	PASS	matches expected limit
limit:rigid blades → no bend	PASS	matches expected limit
knockout:rigid blades	PASS	fast_double -100%, J1_Ns +138%, J2_Ns -100%, gap12_ms -100%, n_hits -50%, defl_lag_H_mm -93.8%, set_lag_H_mm -100%, defl_flap_H_mm -93.6%, defl_fla...
knockout:no blade plasticity	PASS	fast_double -100%, J1_Ns +131%, J2_Ns -100%, gap12_ms -100%, n_hits -50%, defl_lag_H_mm -55.5%, set_lag_H_mm -100%, defl_flap_H_mm -49.2%, defl_fla...
knockout:no blade friction	PASS	fast_double -100%, J1_Ns -6.36%, J2_Ns -100%, gap12_ms -100%, n_hits -50%, defl_lag_H_mm -15.4%, set_lag_H_mm -17.5%, defl_flap_H_mm -16.1%, defl_f...
knockout:no gyroscopic coupling	PASS	fast_double +0%, J1_Ns +0.00934%, J2_Ns +0.128%, gap12_ms +0.126%, n_hits +0%, defl_lag_H_mm +0.00474%, set_lag_H_mm +0.00538%, defl_flap_H_mm -0.0...
sensitivity	PASS	9 inputs swept low/high (trends reported, not judged)
monte_carlo	PASS	20 samples, 0 failed finite/bounds/ledger (0.0%)